

Link Layer Protocols

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Book chapters

Forouzan: 11.6, 11.7

Kihl: 4.4, 4.5

High-level Data Link Control (HDLC)

HDLC is a bit-oriented link protocol.

Frame format:



Flag = 01111110

16- or 32-bit CRC

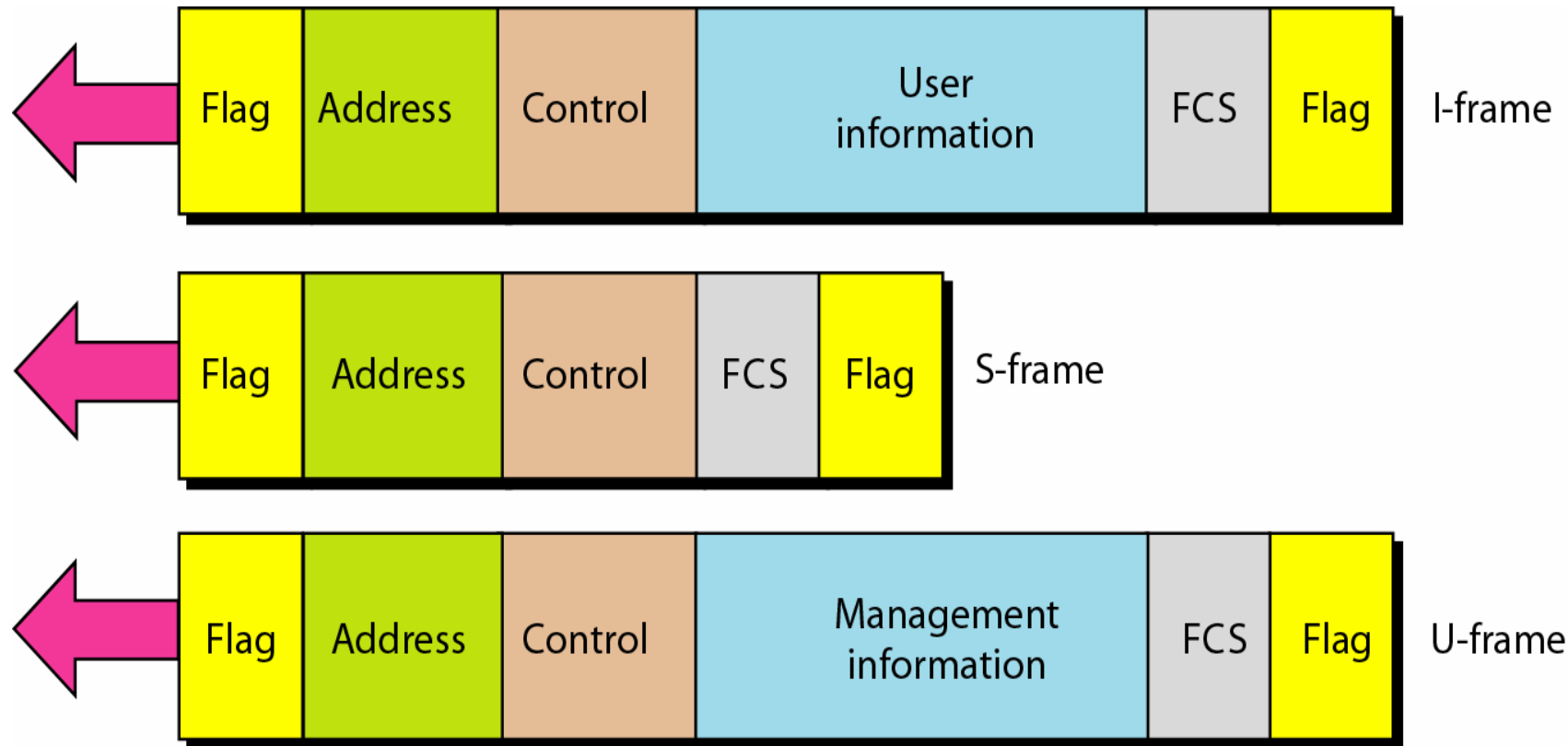
Uses Go-back-N ARQ and Selective Repeat ARQ

HDLC Frames

HDLC defines three types of frames:

- Information frames (**I-frames**) are used to transport user data and control information relating to user data (so called **piggybacking**).
- Supervisory frames (**S-frames**) are used only to transport control information.
- Unnumbered frames (**U-frames**) are intended for managing the link itself.

Frame formats

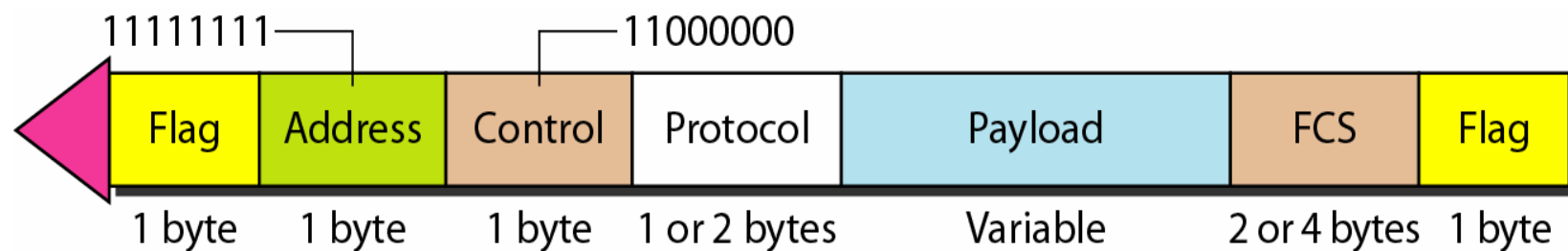


Point-to-Point Protocol (PPP)

PPP is the most common link protocol for modem users.

It is a **byte-oriented** (character-oriented) protocol, that is, everything in the frame is treated in bytes.

The frame format is similar to HDLC:



Byte stuffing

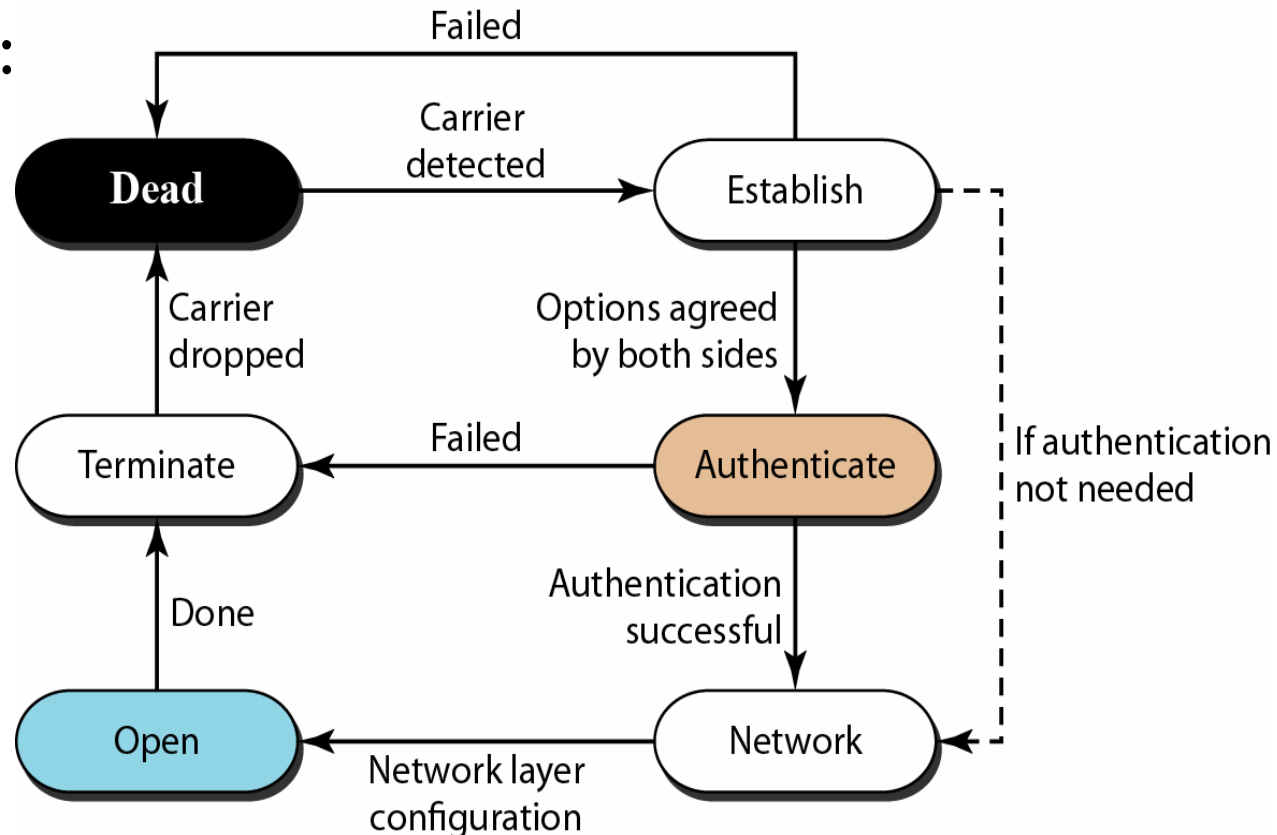
When the flag appears in the data field, it needs to be escaped.

Since PPP is a byte-oriented protocol, **byte stuffing** is used.

Every time the flaglike pattern appears in the data field, the escape byte 01111101 is stuffed to tell the receiver that the next byte is not a flag.

Transition phases

A PPP connection goes through a number of phases:

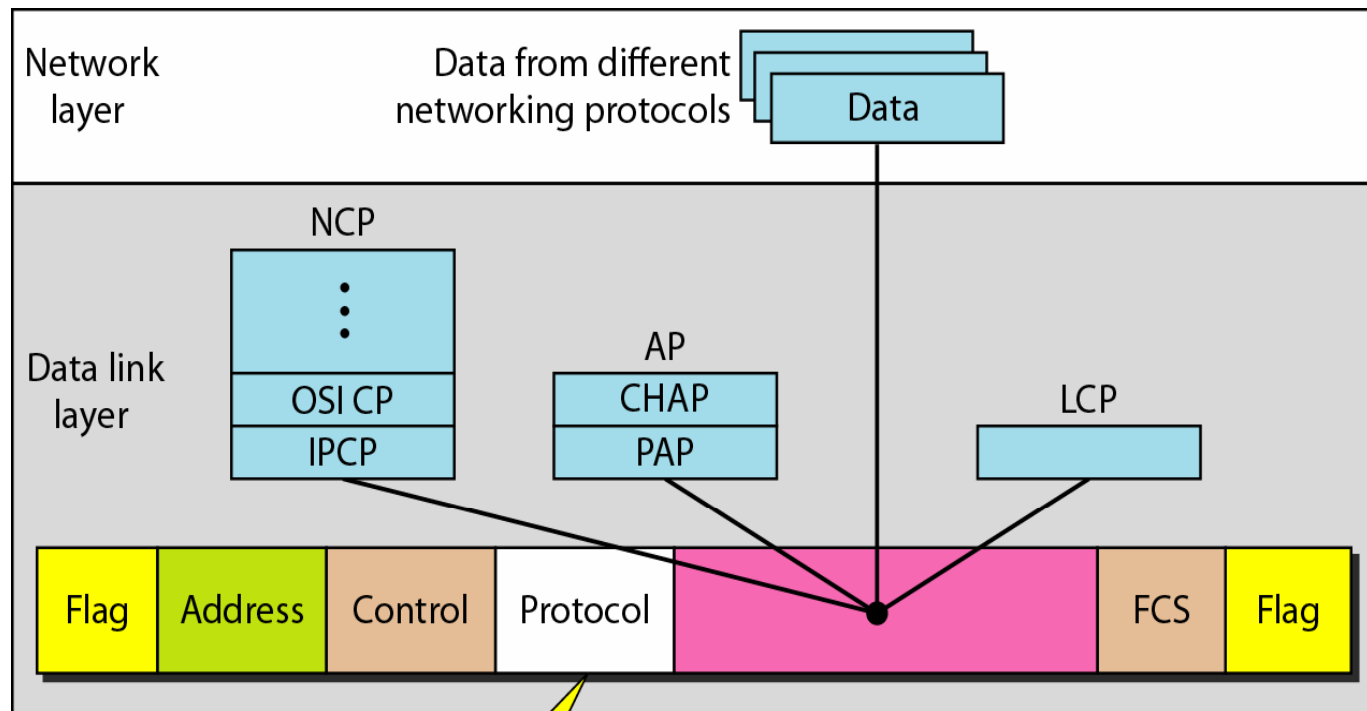


Help protocols

PPP uses a set of other protocols to manage the connection:

- **Link Control Protocol (LCP)** is responsible for establishing, maintaining, configuring, and terminating links.
- **Password Authentication Protocol (PAP)** and **Challenge Handshake Authentication Protocol (CHAP)** are used in the authentication process.
- A **Network Control Protocol (NCP)** configures the link for the network protocol. One example is **Internet Protocol Control Protocol (IPCP)** for Internet.

Other protocols in PPP



LCP: 0xC021
AP: 0xC023 and 0xC223
NCP: 0x8021 and
Data: 0x0021 and

LCP: Link Control Protocol
AP: Authentication Protocol
NCP: Network Control Protocol